A Study On Frames

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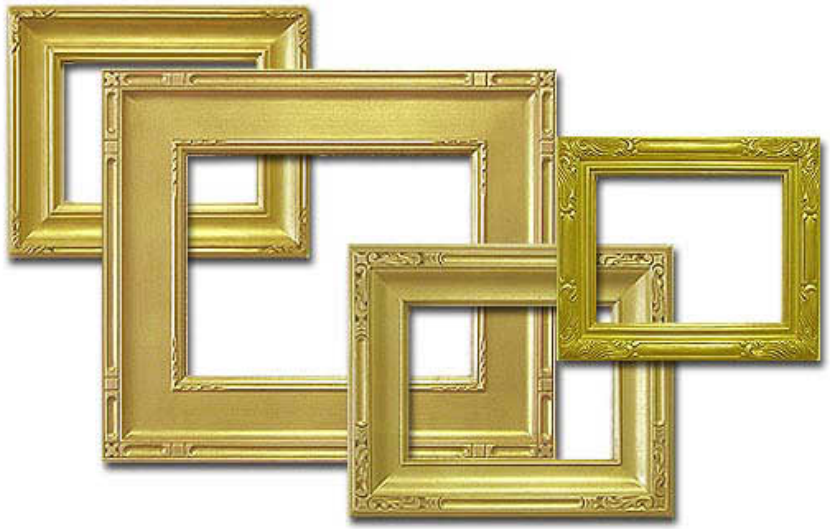
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Temple Graduate Student Seminar, October 2017



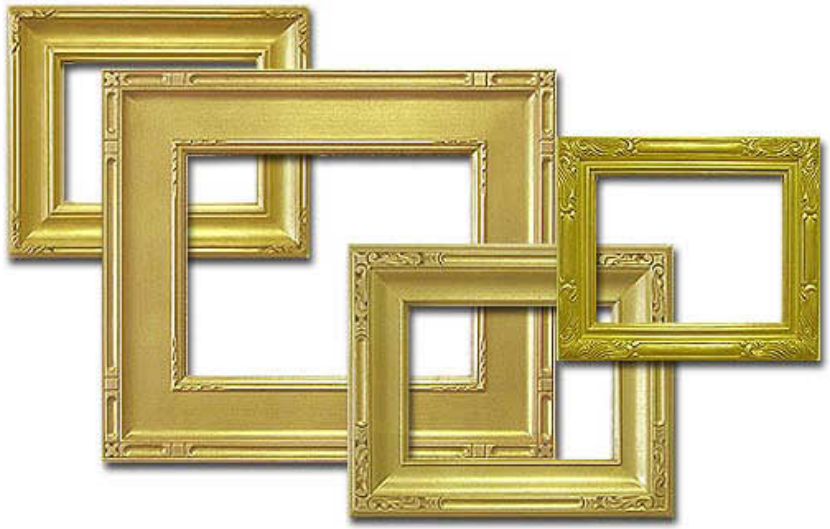
Outline

1. Preliminaries
   - Linear Algebra Review
   - Brief History on Frames and Applications
   - What are Frames?
2. Problem History and Motivation
   - The Kadison-Singer Problem
   - MSS and Example
3. Grassmannian Frames
   - Construction & Example
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A Review of Bases

Definition [12]: Basis

A linearly independent spanning set for a vector space $\mathcal{V}$ is called a **basis** for $\mathcal{V}$.

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The *standard* basis vectors in $\mathbb{R}^3$ are defined as

$$\mathcal{S} = \{e_1, e_2, e_3\} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

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- $\mathcal{S}$ spans $\mathbb{R}^3$ e.g. $u = \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix} = 3e_1 + 4e_2 + 5e_3$
- Linearly independent:

$$c_1 e_1 + c_2 e_2 + c_3 e_3 = 0 \Leftrightarrow \{c_1, c_2, c_3\} = \{0, 0, 0\}$$

Eigenvalues and Eigenvectors of a Matrix

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For an $n \times n$ matrix $\mathbf{M}$, scalars λ and nonzero vectors $v \in \mathbb{C}^n$ satisfying $\mathbf{M}v = \lambda v$ are called *eigenvalues* and *eigenvectors* of $\mathbf{M}$.

The eigenvalues are obtained by solving the *characteristic* equation

$$\det(\mathbf{M} - \lambda \mathbf{I}) = 0.$$

Types of Norms

Our research utilizes two types of *norms*:

- The Euclidean vector norm for a vector $x \in \mathbb{C}^n$ defined as $\|x\| = \sqrt{\langle x, x \rangle}$.
- The Induced matrix norm.

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Other norms which go beyond the scope of this research are

- The Matrix 2-norm, $\|\mathbf{M}\|_2 = \sqrt{\lambda_{max}}$, the max eigenvalue of $\mathbf{M}^*\mathbf{M}$
- The Matrix 1-norm, $\|\mathbf{M}\|_1 =$ the largest absolute column sum.
- The Matrix ∞-norm, $\|\mathbf{M}\|_\infty =$ the largest absolute row sum.
- The Frobenius norm, $\|\mathbf{M}\|_F = \sqrt{trace(\mathbf{M}^*\mathbf{M})}$

Matrix Norms

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Calculating the Norm of a Hermitian Matrix

Given the real matrix $\mathbf{M} = \begin{bmatrix} 1 & \sqrt{2} \\ \sqrt{2} & 2 \end{bmatrix}$, then find $\|\mathbf{M}\|$.

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Notice that $\mathbf{M} = \mathbf{M}^T$. We start by finding eigenvalues of $\mathbf{M}$,

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Therefore, $\|\mathbf{M}\| = 3$. The same procedure applies to complex matrices.

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Induced Matrix Norms

A vector norm that is defined on $\mathbb{C}^p$ for $p = m, n$ *induces* a matrix norm on $\mathbb{C}^{m\times n}$ by setting

$$\|\mathbf{A}\| = \max_{\|x\|=1} \|\mathbf{A}x\|$$

for $\mathbf{A} \in \mathbb{C}^{m\times n}, x \in \mathbb{C}^n$.

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- Note: $\sigma(\mathbf{A})$, the spectrum is the set of all eigenvalues of the matrix $\mathbf{A}$.

Matrix Decomposition

Spectral Decomposition

Given a normal matrix $\mathbf{A} \in \mathbb{C}^{n \times n}$ that has an orthogonal eigenvector basis, it may be decomposed as

$$\mathbf{A} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^*$$

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- A unitary matrix: $\mathbf{U}\mathbf{U}^* = \mathbf{U}^*\mathbf{U} = \mathbf{I}$

Tensor Products

The Tensor Product

Given two vectors $x, y \in \mathbb{C}^n$ the tensor product (outer product) of two vectors is defined as

$$x \otimes y = xy^* = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \begin{bmatrix} \overline{y}_1 & \overline{y}_2 & \cdots & \overline{y}_n \end{bmatrix}$$

$$= \begin{bmatrix} x_1\overline{y}_1 & x_1\overline{y}_2 & \cdots & x_1\overline{y}_n \\ x_2\overline{y}_1 & x_2\overline{y}_2 & \cdots & x_2\overline{y}_n \\ \vdots & \vdots & \ddots & \vdots \\ x_n\overline{y}_1 & x_n\overline{y}_2 & \cdots & x_n\overline{y}_n \end{bmatrix}$$

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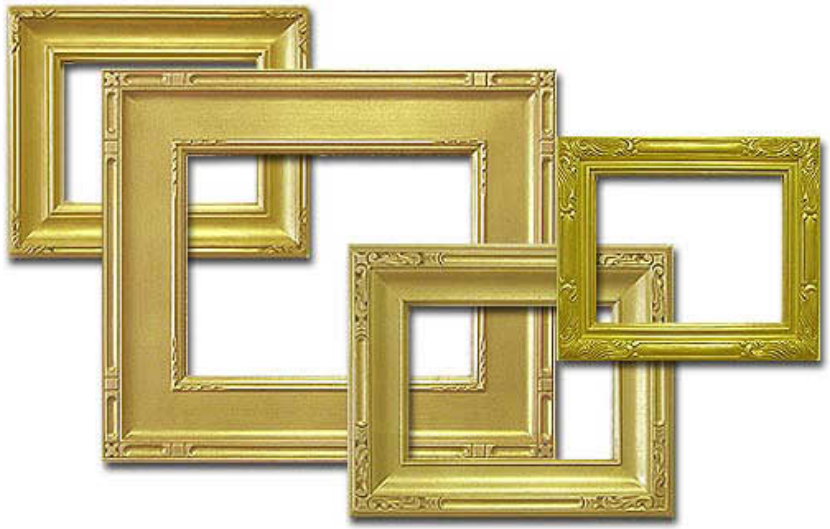
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If $x, y \in \mathbb{C}^n$ then the analogous computation would be $xy^* = x \otimes y$.

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Finite frames [13] were formally defined by Duffin and Schaeffer in 1952.

Developed signal processing further [8, 7] by Daubechies, Grossman, and Meyer in 1986.

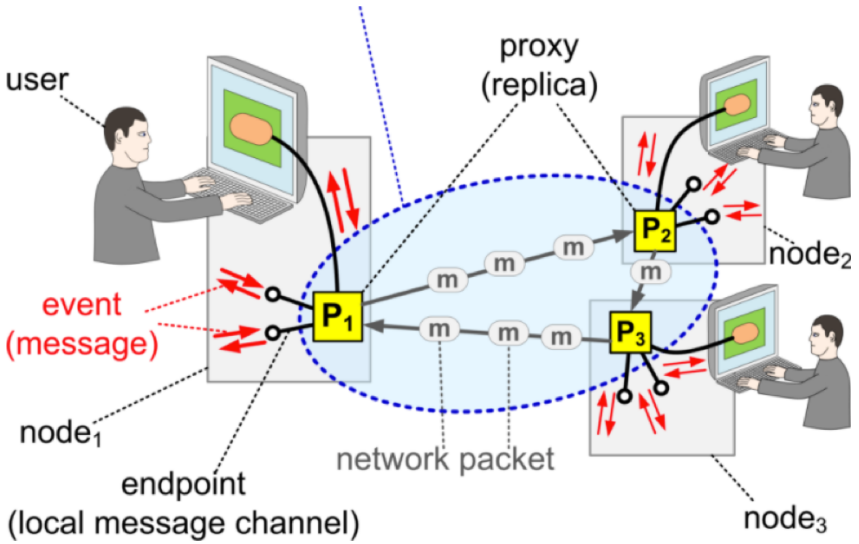
Applications

Sampling Theory [9]: reconstruction of a function from samples i.e. image reconstruction.

Encoding Analog Data [3].

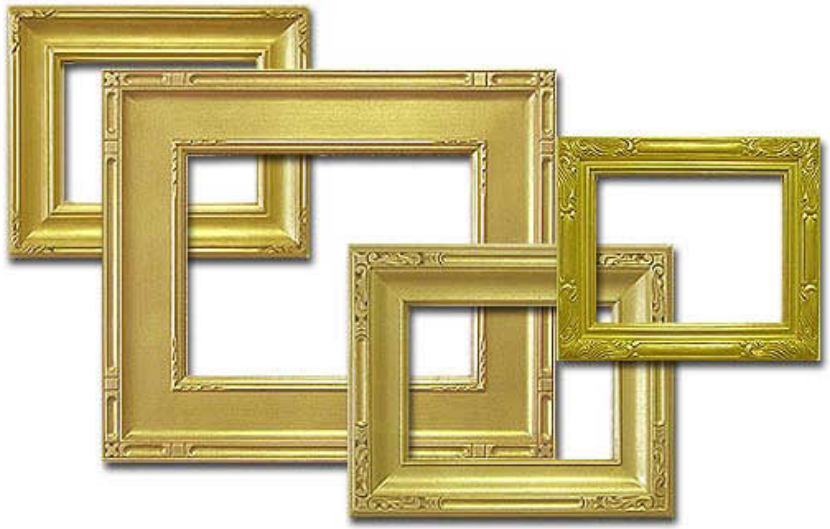
Signal processing of large quantities of redundant data on low dimensional spaces [3].

Fusion Frames [3, 6, 4, 5] used to model distributed processes.



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The Notion of a Frame

Ways to think of a frame...

- a redundant bases [3],
- a collection of vectors that span $\mathbb{H}^n$ $(\mathbb{R}$ or $\mathbb{C})$, not necessarily linearly independent [3],
- a spanning set that has *extra* vectors.

Lemma

A collection of vectors in $\mathbb{H}^n$ is an $\mathbb{H}^n$-frame if and only if it is a spanning set of $\mathbb{H}^n$. [9]

Formal Definition

Definition: Finite Frame [9, 6, 4, 14, 3]

A frame for a complex space $\mathbb{C}^n$ is a sequence of vectors $\{f_i\} \subset \mathbb{C}^n$ for which there exist constants $0 < A \leqslant B < \infty$ such that, for every $x \in \mathbb{C}^n$,

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- This definition can be extended to Hilbert spaces $\mathcal{H}$.

Crash Course With Finite Frames

Example

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- but the vectors are not linearly independent.
- Notice that the vectors are not all necessarily orthonormal, but we can form frames from orthonormal vector sets.

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To do this we will form a matrix $\mathbf{T}$ where the columns of $\mathbf{T}$ are the frame vectors from $\mathcal{F}$

$$\mathbf{T} = \begin{bmatrix} \uparrow & \uparrow & \uparrow & \uparrow & \uparrow \\ f_1 & f_2 & f_3 & f_4 & f_5 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \end{bmatrix} = \begin{bmatrix} 1 & 2 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

The $\mathbf{T}$ matrix is called the *synthesis* operator of the frame.

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We need to find frame bounds A and B!

To do this we will form a matrix $\mathbf{T}$ where the columns of $\mathbf{T}$ are the frame vectors from $\mathcal{F}$

$$\mathbf{T} = \begin{bmatrix} \uparrow & \uparrow & \uparrow & \uparrow & \uparrow \\ f_1 & f_2 & f_3 & f_4 & f_5 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \end{bmatrix} = \begin{bmatrix} 1 & 2 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

The $\mathbf{T}$ matrix is called the *synthesis* operator of the frame.

Next we find $\mathbf{S} = \mathbf{T}\mathbf{T}^*$, this is called the *frame* operator.

Crash Course With Finite Frames

Example

If we find the eigenvalues of **S** we get the following

$$\lambda_1 = 6.19258...$$
$$\lambda_2 = 2$$
$$\lambda_3 = 0.807418...$$

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$$\lambda_1 = 6.19258...$$
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The lower frame bound coincides with the smallest eigenvalue
$\Rightarrow A = 0.807418...$

The upper frame bound conincides with the largest eigenvalue
$\Rightarrow B = 6.19258...$

Equiangular Tight Frames

Definition: Equiangular Tight Frames (ETF)

A family of vectors $\{f_i\}_{i=1}^m$ in $\mathbb{H}^n$ that satisfy the following conditions:

1. $\|f_i\|_2 = 1$ for all $i \in \{1, \ldots, m\}$,
2. $|\langle f_i, f_j \rangle| = c$ for all $i, j \in \{1, \ldots, m\}$ and $i \neq j$,
3. $\frac{n}{m} \sum_{i=1}^m \langle f, f_i \rangle f_i = f$ for all $f \in \mathbb{H}^n$;

is known as an equiangular tight frame.

If a frame meets condition (1) it is referred to as an *unit-norm* frame.

If a frame meets condition (2) is is referred to as an *equiangular* frame.

If $\|f_i\|_2$ is constant for every frame vector, then it is called *equinorm*.

The Mercedes-Benz Frame

Example

Let $m = 3$ and we have the following equinorm-equiangular frame:

$$\mathcal{F} = \{f_0, f_1, f_2\} = \left\{ \begin{bmatrix} \sqrt{\frac{2}{3}} \\ 0 \end{bmatrix}, \begin{bmatrix} -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}, \begin{bmatrix} -\frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix} \right\}.$$

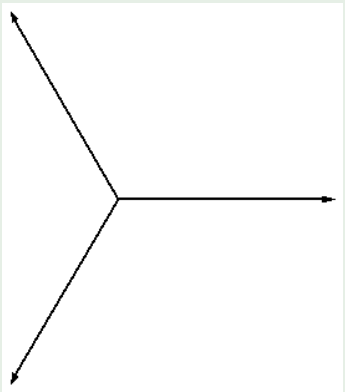
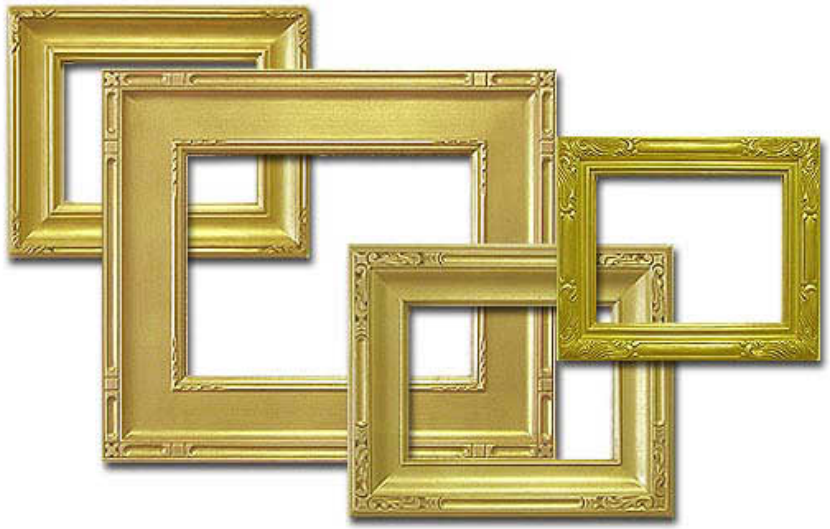


Figure: Mercedes-Benz Frame

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KS Conjecture

We explore an extension to the Kadison-Singer Problem (KS) involving special vector sets and partitioning algorithms.

In 1959 Richard Kadison and Isadore Singer posed the following problem [10, 11, 2, 15, 1]:

Kadison-Singer Problem (KS)

Does every pure state on the (abelian) von Neumann algebra $\mathbb{D}$ of bounded diagonal operators in ℓ_2 have a unique extension to a (pure) state on $B(\ell_2)$, the von Neumann algebra of all bounded linear operators on the Hilbert space ℓ_2?

Originally Kadison thought the answer to this question was negative.

Motivation

2015: conjecture proven true by Marcus, Spielman, and Srivastava [11].

Expanded upon previous work by [15, 1] and formulated *Corollary 1.5*.

> *...can one design an efficient algorithm to find the partitions and pavings that are guaranteed to exist by Corollary 1.5 (MSS Theorem).*

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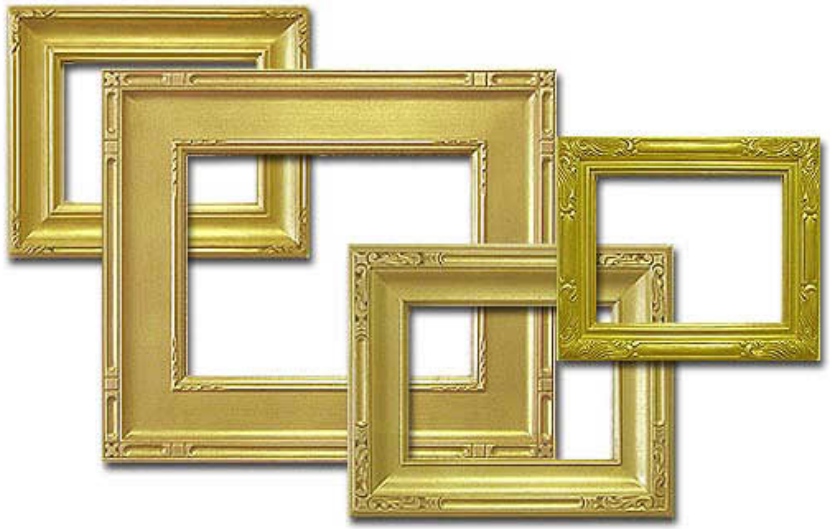
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Relevant in the fields of Computer Science, Computational Mathematics, and Signal Processing we are always in need of finding algorithms to partition data effectively.

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MSS Theorem (*Corollary 1.5*)

Let r be a positive integer and $u_1, ..., u_m \in \mathbb{C}^d$ be vectors such that

$$\sum_{i=1}^{m} u_i \otimes u_i = \mathbf{I}$$

and $\|u_i\|^2 \leqslant \delta$ for all i. Then there exists a partition $\{S_1, ..., S_r\}$ of $[m]$ such that

$$\left\| \sum_{i \in S_j} u_i \otimes u_i \right\| \leqslant \left(\frac{1}{\sqrt{r}} + \sqrt{\delta} \right)^2 \qquad \text{(MSS)}$$

for $j = 1, ..., r$.

Notice: $[m] = S_1 \cup S_2 \cup \cdots \cup S_r$

A partition $\{S_1, ..., S_r\}$ that satisfies the MSS inequality will be called a *valid* partition.

An Example of the MSS Theorem

Example

Suppose we have the following three vectors (these form the *Mercedes-Benz* frame)

$$\mathcal{U} = \{u_1, u_2, u_3\} = \left\{ \begin{bmatrix} \sqrt{\frac{2}{3}} \\ 0 \end{bmatrix}, \begin{bmatrix} -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}, \begin{bmatrix} -\frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix} \right\}.$$

- We have $\sum_{i=1}^{3} u_i \otimes u_i = \mathbf{I}$.
- And $\|u_i\|^2 = \frac{2}{3}$, so $\delta = \frac{2}{3}$.

Therefore, the three given vectors u_1, u_2, u_3 satisfy the hypothesis of the MSS Theorem.

An Example of the MSS Theorem

Example

The MSS Theorem then states there exists a partition.
Let us find a partition!

- We only have three vectors; hence $[m] = \{1, 2, 3\}$
- Choose the number of subsets in the partitions $r = 2$
- A simple 2 subset partition: let $S_1 = \{1\}$ and $S_2 = \{2, 3\}$

$$S_1 \Rightarrow \mathcal{P}_1 = \{u_1\} = \left\{ \begin{bmatrix} \sqrt{\frac{2}{3}} \\ 0 \end{bmatrix} \right\}$$

$$S_2 \Rightarrow \mathcal{P}_2 = \{u_2, u_3\} = \left\{ \begin{bmatrix} -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}, \begin{bmatrix} -\frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix} \right\}$$

An Example of the MSS Theorem

Example

- We have $r = 2$ and $\delta = \frac{2}{3}$; therefore the bound described by the MSS inequality is

$$\left(\frac{1}{\sqrt{r}} + \sqrt{\delta}\right)^2 = \left(\frac{1}{\sqrt{2}} + \sqrt{\frac{2}{3}}\right)^2 = \frac{7}{6} + \frac{2}{\sqrt{3}} > 1$$

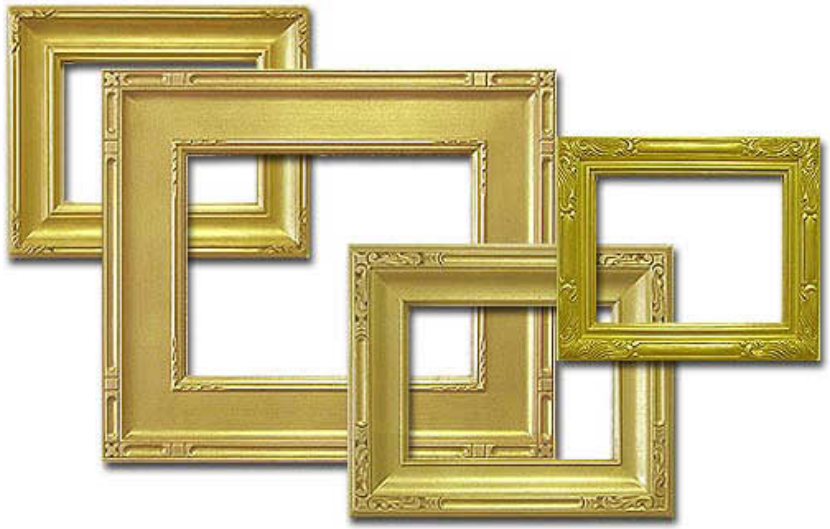
- The corresponding norms of the tensor product sums for $\mathcal{P}_1, \mathcal{P}_2$ are

$$\|u_1 \otimes u_1\| = \frac{2}{3}$$
$$\|u_2 \otimes u_2 + u_3 \otimes u_3\| = 1$$

Both of these norms are less than the bound $\frac{7}{6} + \frac{2}{\sqrt{3}}$.
$\Rightarrow \{S_1, S_2\}$ is a *valid* partition of $\mathcal{U}$.

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Gassmannian Frames

- Class of frames generated by skew symmetric conference matrices:

Definition

A matrix $\mathbf{C}$ of order n with diagonal elements 0 and off diagonal elements ± 1 satisfying

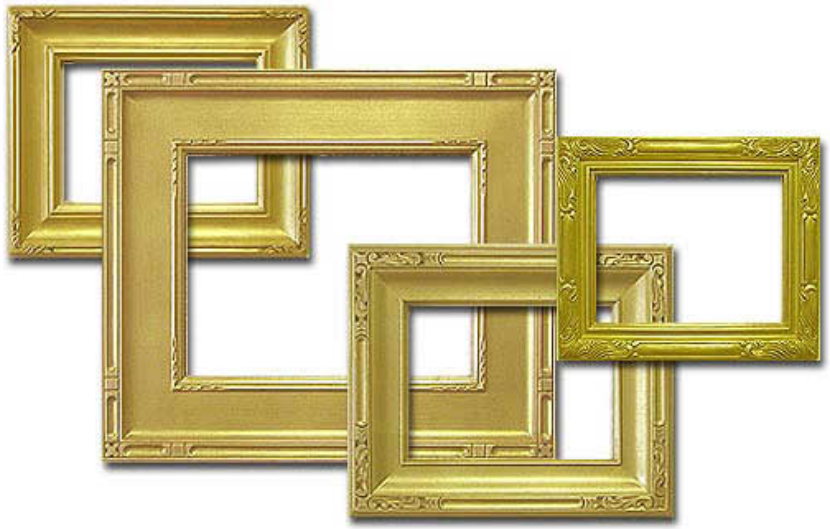
$$\mathbf{C}\mathbf{C}^T = (n-1)\mathbf{I}$$

is said to be a conference matrix.

- Strohmer & Heath [14] give a construction: $\mathbf{R} = \mathbf{I} + i\alpha\mathbf{C}$.
- $\alpha = \sqrt{\frac{n-m}{m(n-1)}}$, in general $n \geqslant m$, for *optimal* case $m = n/2$.
- Recursive construction beginning with 2×2 conference matrix.
- Then find spectral decomposition of $\mathbf{R}$: $\mathbf{R} = \mathbf{W}\mathbf{\Lambda}\mathbf{W}^*$
- The frame vectors are in the rows of $\mathbf{W}$

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Frame Correlation and the Grassmannian

Example

Suppose we have the following set of vectors

$$\mathcal{F} = \{f_j\}_{j=1}^{4} = \left\{ \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}, \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}, \begin{bmatrix} \frac{1}{2} \\ -\frac{\sqrt{3}}{2} \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \end{bmatrix} \right\}$$

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- $\mathcal{F}$ is a frame because the vectors span $\mathbb{R}^2$.

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- $\mathcal{F}$ is a frame because the vectors span $\mathbb{R}^2$.
- Notice that that vectors are not all linearly independent of each other,
- and the norm of each vector is 1.

Frame Correlation and the Grassmannian

Example

Next we calculate the magnitude of every possible inner product, with the exception of a vector with itself.

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Next we calculate the magnitude of every possible inner product, with the exception of a vector with itself.

$$|\langle f_1, f_2 \rangle| = 1$$
$$|\langle f_1, f_3 \rangle| = 0.258819...$$
$$|\langle f_1, f_4 \rangle| = 0.707107...$$
$$|\langle f_2, f_3 \rangle| = 0.258819...$$
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$$|\langle f_3, f_4 \rangle| = 0.5$$

The largest value is called the *maximal frame correlation* $\mathcal{M}(\mathcal{F}) = 1$.

Frame Correlation and the Grassmannian

Definition: Maximal Frame Correlation

For a given *unit norm frame* $\mathcal{F} = \{f_k\}_{k=1}^n$ in $\mathbb{E}^m$ we define the maximal frame correlation $\mathcal{M}(\mathcal{F})$ by

$$\mathcal{M}(\mathcal{F}) = \max_{k,l,k\neq l} \{|\langle f_k, f_l \rangle|\}$$

Frame Correlation and the Grassmannian

Definition: Grassmannian Frame

Given all unit norm frames $\mathcal{F}$ in $\mathbb{E}^m$ we denote

$$\mu = \min\{\mathcal{M}(\mathcal{F})\}.$$

Where μ is the *smallest* of the maximal frame correlations.

A set of vectors $\mathcal{U} = \{u_1, ..., u_n\}$ is a Grassmannian frame if

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- A Grassmannian frame has the *smallest* maximal frame correlation taken over all unit norm frames.
- It is known that $\mu \geqslant \sqrt{\frac{n-m}{m(n-1)}}$.
- We will be analyzing *optimal* Grassmannian frames, this occurs when equality holds.

One More Definition

Definition (Subset Norm)

Let $\mathcal{F} = \{f_1, \ldots, f_n\}$ be a set of vectors and $\mathcal{S}$ a subset of $[n] = \{1, \ldots n\}$. We define the *subset outer product* of $\mathcal{S}$ to be matrix

$$\mathbf{F}_{\mathcal{S}} = \sum_{j \in \mathcal{S}} f_j \otimes f_j$$

and the *subset norm* of $\mathcal{S}$ to be the induced matrix (spectral) norm $\|\mathbf{F}_{\mathcal{S}}\|$.

Example: $n = 16$

We calculate $\mathbf{R} =$

$$\begin{pmatrix}
1 & -\frac{i}{\sqrt{15}} & -\frac{i}{\sqrt{15}} & -\frac{i}{\sqrt{15}} & -\frac{i}{\sqrt{15}} & -\frac{i}{\sqrt{15}} & -\frac{i}{\sqrt{15}} & -\frac{i}{\sqrt{15}} & -\frac{i}{\sqrt{15}} & -\frac{i}{\sqrt{15}} & -\frac{i}{\sqrt{15}} & -\frac{i}{\sqrt{15}} & -\frac{i}{\sqrt{15}} & -\frac{i}{\sqrt{15}} & -\frac{i}{\sqrt{15}} & -\frac{i}{\sqrt{15}} \\
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\frac{i}{\sqrt{15}} & -\frac{i}{\sqrt{15}} & -\frac{i}{\sqrt{15}} & -\frac{i}{\sqrt{15}} & -\frac{i}{\sqrt{15}} & -\frac{i}{\sqrt{15}} & -\frac{i}{\sqrt{15}} & -\frac{i}{\sqrt{15}} & -\frac{i}{\sqrt{15}} & -\frac{i}{\sqrt{15}} & \frac{i}{\sqrt{15}} & \frac{i}{\sqrt{15}} & 1 & \frac{i}{\sqrt{15}} & -\frac{i}{\sqrt{15}} & -\frac{i}{\sqrt{15}} \\
\frac{i}{\sqrt{15}} & \frac{i}{\sqrt{15}} & \frac{i}{\sqrt{15}} & \frac{i}{\sqrt{15}} & \frac{i}{\sqrt{15}} & \frac{i}{\sqrt{15}} & \frac{i}{\sqrt{15}} & \frac{i}{\sqrt{15}} & \frac{i}{\sqrt{15}} & \frac{i}{\sqrt{15}} & -\frac{i}{\sqrt{15}} & -\frac{i}{\sqrt{15}} & \frac{i}{\sqrt{15}} & 1 & \frac{i}{\sqrt{15}} & -\frac{i}{\sqrt{15}} \\
\frac{i}{\sqrt{15}} & -\frac{i}{\sqrt{15}} & -\frac{i}{\sqrt{15}} & -\frac{i}{\sqrt{15}} & -\frac{i}{\sqrt{15}} & -\frac{i}{\sqrt{15}} & -\frac{i}{\sqrt{15}} & -\frac{i}{\sqrt{15}} & -\frac{i}{\sqrt{15}} & -\frac{i}{\sqrt{15}} & \frac{i}{\sqrt{15}} & \frac{i}{\sqrt{15}} & -\frac{i}{\sqrt{15}} & -\frac{i}{\sqrt{15}} & 1 & \frac{i}{\sqrt{15}} \\
\frac{i}{\sqrt{15}} & \frac{i}{\sqrt{15}} & -\frac{i}{\sqrt{15}} & \frac{i}{\sqrt{15}} & \frac{i}{\sqrt{15}} & \frac{i}{\sqrt{15}} & \frac{i}{\sqrt{15}} & \frac{i}{\sqrt{15}} & \frac{i}{\sqrt{15}} & \frac{i}{\sqrt{15}} & -\frac{i}{\sqrt{15}} & -\frac{i}{\sqrt{15}} & \frac{i}{\sqrt{15}} & \frac{i}{\sqrt{15}} & -\frac{i}{\sqrt{15}} & 1
\end{pmatrix}$$

Example: $n = 16$

Here are the frame vectors

$$\begin{pmatrix}
\frac{i}{4}-\frac{1}{4\sqrt{15}} & \frac{-15+7i\sqrt{15}}{30\sqrt{14}} & \frac{i\left(i+\sqrt{15}\right)}{2\sqrt{70}} & \frac{5+3i\sqrt{15}}{20\sqrt{3}} & \frac{i\left(i+\sqrt{15}\right)}{6\sqrt{10}} & \frac{4i}{3\sqrt{35}} & \frac{i\left(i+\sqrt{15}\right)}{2\sqrt{105}} & \frac{i}{\sqrt{30}} \\
\frac{i}{4}-\frac{1}{4\sqrt{15}} & -2i\sqrt{\frac{2}{105}} & \frac{1}{3}\sqrt{\frac{2}{35}}\left(1+i\sqrt{15}\right) & \frac{15-7i\sqrt{15}}{60\sqrt{3}} & \frac{i\left(i+\sqrt{15}\right)}{6\sqrt{10}} & \frac{15-7i\sqrt{15}}{30\sqrt{21}} & \frac{1+i\sqrt{15}}{2\sqrt{105}} & -\frac{i}{\sqrt{30}} \\
\frac{1}{60}\left(-15i+\sqrt{15}\right) & 2i\sqrt{\frac{2}{105}} & \frac{-11+3i\sqrt{15}}{6\sqrt{70}} & \frac{15-7i\sqrt{15}}{60\sqrt{3}} & \frac{1-i\sqrt{15}}{6\sqrt{10}} & \frac{15+7i\sqrt{15}}{30\sqrt{21}} & \frac{1+i\sqrt{15}}{2\sqrt{105}} & -\frac{i}{\sqrt{30}} \\
\frac{i}{4}-\frac{1}{4\sqrt{15}} & \frac{-15+7i\sqrt{15}}{30\sqrt{14}} & \frac{1}{3}\sqrt{\frac{2}{35}}\left(1-i\sqrt{15}\right) & \frac{15-7i\sqrt{15}}{60\sqrt{3}} & \frac{3+i\sqrt{15}}{6\sqrt{10}} & \frac{i}{\sqrt{35}} & \frac{1-i\sqrt{15}}{2\sqrt{105}} & -\frac{i}{\sqrt{30}} \\
\frac{1}{60}\left(-15i+\sqrt{15}\right) & \frac{15-7i\sqrt{15}}{30\sqrt{14}} & \frac{1-i\sqrt{15}}{2\sqrt{70}} & \frac{5+i\sqrt{15}}{10\sqrt{3}} & \frac{1+i\sqrt{15}}{6\sqrt{10}} & \frac{15+7i\sqrt{15}}{30\sqrt{21}} & \frac{1+i\sqrt{15}}{2\sqrt{105}} & -\frac{i}{\sqrt{30}} \\
\frac{1}{60}\left(-15i+\sqrt{15}\right) & 2i\sqrt{\frac{2}{105}} & \frac{17+3i\sqrt{15}}{6\sqrt{70}} & \frac{i}{3\sqrt{5}} & \frac{1+i\sqrt{15}}{6\sqrt{10}} & -\frac{4i}{3\sqrt{35}} & \frac{1-i\sqrt{15}}{2\sqrt{105}} & -\frac{i}{\sqrt{30}} \\
\frac{i}{4}-\frac{1}{4\sqrt{15}} & \frac{45+7i\sqrt{15}}{30\sqrt{14}} & \frac{1-i\sqrt{15}}{3\sqrt{70}} & \frac{i}{3\sqrt{5}} & -\frac{i\left(-i+\sqrt{15}\right)}{6\sqrt{10}} & \frac{15-7i\sqrt{15}}{30\sqrt{21}} & \frac{1+i\sqrt{15}}{2\sqrt{105}} & -\frac{i}{\sqrt{30}} \\
\frac{i}{4}+\frac{7}{4\sqrt{15}} & -i\sqrt{\frac{2}{105}} & \frac{1+i\sqrt{15}}{3\sqrt{70}} & \frac{i}{3\sqrt{5}} & \frac{3-i\sqrt{15}}{6\sqrt{10}} & \frac{i}{\sqrt{35}} & \frac{1-i\sqrt{15}}{2\sqrt{105}} & -\frac{i}{\sqrt{30}} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{\sqrt{2}} \\
0 & 0 & 0 & 0 & 0 & 0 & \sqrt{\frac{7}{15}} & \frac{i}{\sqrt{30}} \\
0 & 0 & 0 & 0 & 0 & \sqrt{\frac{3}{7}} & -\frac{i\left(-i+\sqrt{15}\right)}{2\sqrt{105}} & \frac{i}{\sqrt{30}} \\
0 & 0 & 0 & 0 & \sqrt{\frac{2}{5}} & -\frac{i}{\sqrt{35}} & \frac{i\left(i+\sqrt{15}\right)}{2\sqrt{105}} & \frac{i}{\sqrt{30}} \\
0 & 0 & 0 & \frac{1}{\sqrt{3}} & -\frac{i\left(-i+\sqrt{15}\right)}{6\sqrt{10}} & \frac{-15-7i\sqrt{15}}{30\sqrt{21}} & -\frac{i\left(-i+\sqrt{15}\right)}{2\sqrt{105}} & \frac{i}{\sqrt{30}} \\
0 & 0 & \frac{\sqrt{\frac{14}{5}}}{3} & -\frac{i}{3\sqrt{5}} & -\frac{i\left(-i+\sqrt{15}\right)}{6\sqrt{10}} & \frac{4i}{3\sqrt{35}} & \frac{i\left(i+\sqrt{15}\right)}{2\sqrt{105}} & \frac{i}{\sqrt{30}} \\
0 & \sqrt{\frac{2}{7}} & \frac{i\left(i+\sqrt{15}\right)}{3\sqrt{70}} & -\frac{i}{3\sqrt{5}} & \frac{1+i\sqrt{15}}{6\sqrt{10}} & \frac{-15+7i\sqrt{15}}{30\sqrt{21}} & -\frac{i\left(-i+\sqrt{15}\right)}{2\sqrt{105}} & \frac{i}{\sqrt{30}} \\
\frac{2}{\sqrt{15}} & i\sqrt{\frac{2}{105}} & -\frac{i\left(-i+\sqrt{15}\right)}{3\sqrt{70}} & -\frac{i}{3\sqrt{5}} & \frac{i\left(3i+\sqrt{15}\right)}{6\sqrt{10}} & -\frac{i}{\sqrt{35}} & \frac{i\left(i+\sqrt{15}\right)}{2\sqrt{105}} & \frac{i}{\sqrt{30}}
\end{pmatrix}$$

The rows are the frame vectors, $n = 16$ vectors with dimension $m = 8$.

Norm of the **R**-matrix

Theorem

The norm of the matrix $\mathbf{R} = \mathbf{I} + i\alpha\mathbf{C}$*, where* $\mathbf{C}$ *is a* $n \times n$ *skew symmetric conference matrix of order* $n = 2^k$ *and* α *is a real constant, is given by*

$$\|\mathbf{R}\| = 1 + |\alpha|\sqrt{n-1}.$$

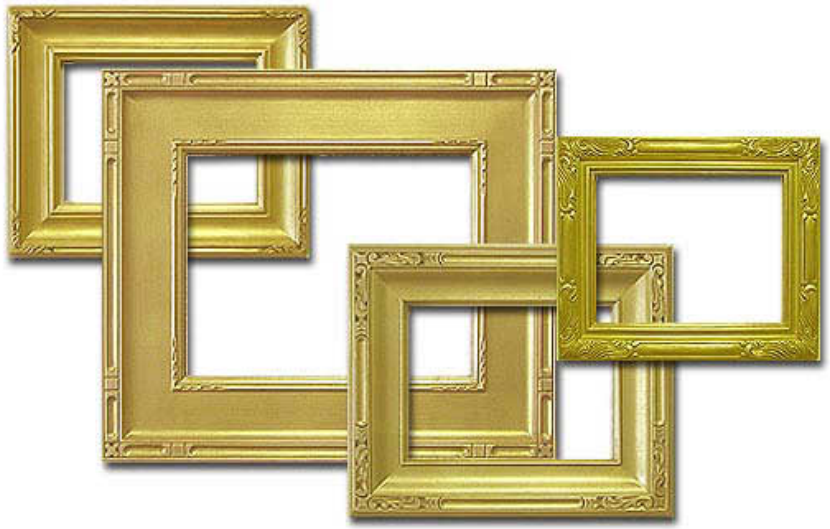
It turns out we can find the norms of particular sub-blocks of $\mathbf{R}$

$$\|\mathbf{F}_{\mathcal{S}_{d,q}}\| = \frac{1}{2} + \frac{1}{2}\frac{\sqrt{2^{k-d}-1}}{\sqrt{2^k-1}} = \frac{1}{2}\|\mathbf{R}_{d,q}(k)\|, \tag{1}$$

where $d \in \{0, \ldots, k-1\}$.

Outline

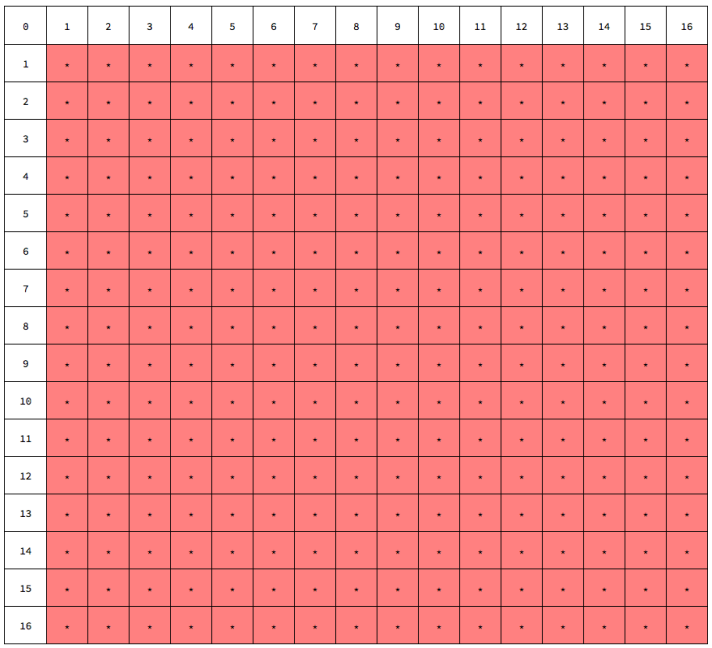
1. Preliminaries
   - Linear Algebra Review
   - Brief History on Frames and Applications
   - What are Frames?
2. Problem History and Motivation
   - The Kadison-Singer Problem
   - MSS and Example
3. Grassmannian Frames
   - Construction & Example
   - What are Grassmannian Frames?
   - Frame Partition Algorithm



Algorithm Example: $\mathcal{F} = \{f_j\}_{j=1}^{16}$

Example

- Construct a 16 vector ETF, $\mathcal{F} = \{f_j\}_{j=1}^{16}$.
- Find an $r = 5$ subset partition $\mathcal{P} = \{\mathcal{S}_1, \ldots, \mathcal{S}_5\}$ of $\mathcal{F}$.

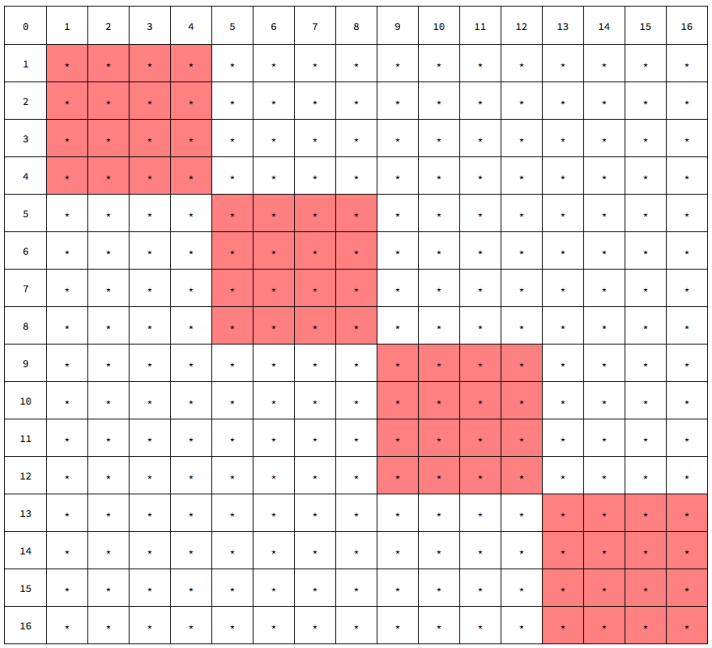


$\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16\}$

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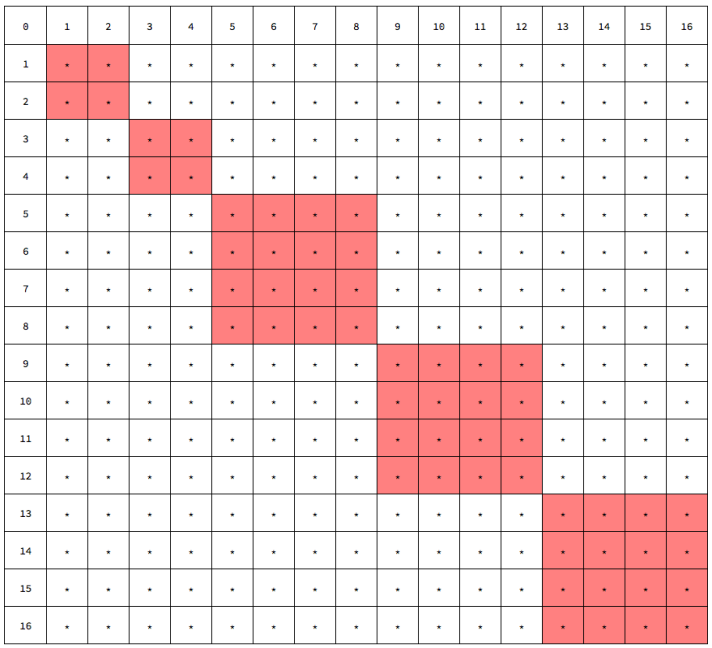
$\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16\}$

$\{1, 2, 3, 4\}, \{5, 6, 7, 8\}, \{9, 10, 11, 12\}, \{13, 14, 15, 16\}$

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$\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16\}$

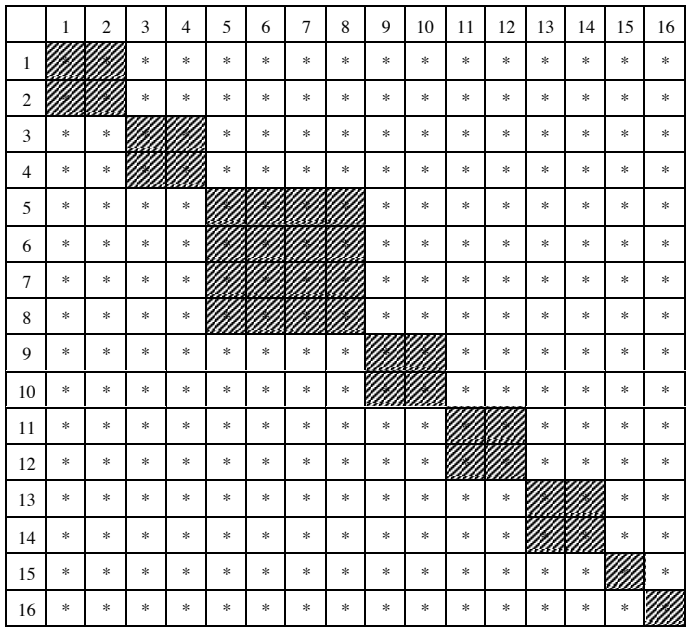
$\{1, 2, 3, 4\}, \{5, 6, 7, 8\}, \{9, 10, 11, 12\}, \{13, 14, 15, 16\}$

$\{1, 2\}, \{3, 4\}, \{5, 6, 7, 8\}, \{9, 10, 11, 12\}, \{13, 14, 15, 16\}$

General Partition Example

Example

- Construct a 16 vector ETF, $\mathcal{F} = \{f_j\}_{j=1}^{16}$.
- Find an $r = 8$ subset partition $\mathcal{P} = \{\mathcal{S}_1, \ldots, \mathcal{S}_8\}$ of $\mathcal{F}$.



Subset	$\|\mathbf{F}_{\mathcal{S}_h}\|$
$\mathcal{S}_1 = \{1, 2\}$	0.629099
$\mathcal{S}_2 = \{3, 4\}$	0.629099
$\mathcal{S}_3 = \{5, 6, 7, 8\}$	0.723607
$\mathcal{S}_4 = \{9, 10\}$	0.629099
$\mathcal{S}_5 = \{11, 12\}$	0.629099
$\mathcal{S}_6 = \{13, 14\}$	0.629099
$\mathcal{S}_7 = \{15\}$	0.5
$\mathcal{S}_8 = \{16\}$	0.5

Table: All norms $< \frac{1}{2} + \frac{1}{\sqrt{2r}} = 0.75$

Algorithm

Algorithm 1: Diagonal Partition Algorithm

Result: $\mathcal{I} = \{1, \ldots, 2^k\}$ is partitioned into $\mathcal{P} = \{\mathcal{S}_{d_i,q_i}\}_{i=1}^{r}$.
Given $r > 1$;
Initialize $d = \lfloor \log_2 r \rfloor$, $r' = 2^d$;
Initialize $\mathcal{P} = \emptyset$;
for $i = 1$ **to** $2(r - r')$ **do**
  $\mathcal{S}_{d+1,i} = \{(i-1) \cdot 2^{k-(d+1)} + 1, \ldots, i \cdot 2^{k-(d+1)}\}$;
  $\mathcal{P} = \mathcal{P} \cup \{\mathcal{S}_{d+1,i}\}$;
end
for $i = r - r' + 1$ **to** r' **do**
  $\mathcal{S}_{d,i} = \{(i-1) \cdot 2^{k-d} + 1, \ldots, i \cdot 2^{k-d}\}$;
  $\mathcal{P} = \mathcal{P} \cup \{\mathcal{S}_{d,i}\}$;
end

- Our algorithm yields subsets with corresponding tensor norm *sharper* then the MSS bound.
- Our results apply to complex ETFs constructed from skew symmetric conference matrices.
- We have shown that partitions containing subsets of size 2 or 3 are MSS partitions.
- We are exploring partitions of ETFs constructed using Steiner systems.

References I



Joel Anderson. "Extensions, Restrictions, and Representations of States on C^*- Algebras". In: *Transactions of the American Mathematical Society* 249.2 (1979), pp. 303–329. ISSN: 00029947.



P. G. Casazza and J. Crandell Tremain. "The Kadison-Singer Problem in mathematics and engineering". In: *Proceedings of the National Academy of Science* 103 (Feb. 2006), pp. 2032–2039.



P. G. Casazza and G. Kutyniok. *Finite Frames: Theory and Applications*. New York, New York: Springer, 2013.



O. Christensen. *An Introduction to Frames and Riesz Bases*. Boston, Massachusetts: Birkhauser, 2003. ISBN: 978-0-8176-4295-2.



O. Christensen. *An Introduction to Frames and Riesz Bases*. Boston, Massachusetts: Birkhauser, 2016. ISBN: 978-3-3192-5611-5.

References II



O. Christensen. *Frames And Bases: An Introductory Course*. Boston, Massachusetts: Birkhauser, 2008. ISBN: 978-0-8176-4677-6.



Ingrid Daubechies and A. Grossmann. "Frames in the bargmann space of entire functions". In: *Communications on Pure and Applied Mathematics* 41.2 (1988), pp. 151–164. ISSN: 1097-0312. DOI: 10.1002/cpa.3160410203. URL: http://dx.doi.org/10.1002/cpa.3160410203.



Ingrid Daubechies, A. Grossmann, and Y. Meyer. "Painless nonorthogonal expansions". In: *Journal of Mathematical Physics* 27.5 (1986), pp. 1271–1283. DOI: 10.1063/1.527388. URL: http://dx.doi.org/10.1063/1.527388.



D. Han et al. *Frames for Undergraduates*. Vol. 40. Providence, Rhode Island: American Mathematical Society, 2007.

References III



A. Marcus, D. A. Spielman, and N. Svrivastava. "Interlacing Families I: Mixed Characteristic Polynomials and the Kadison-Singer Problem.". In: *Annals of Mathematics* 182 (2015), pp. 307–325. DOI: 10.4007/annals.2015.182.1.7.



A. Marcus, D. A. Spielman, and N. Svrivastava. "Interlacing families II: Mixed characteristic polynomials and the Kadison-Singer problem.". In: *Annals of Mathematics* 182 (2015), pp. 327–350. DOI: 10.4007/annals.2015.182.1.8.



C. D. Meyer. *Matrix Analysis and Applied Linear Algebra*. Philadelphia, Pennsylvania: SIAM, 2000.



A. C. Schaeffer R. J. Duffin. "A Class of Nonharmonic Fourier Series". In: *Transactions of the American Mathematical Society* 72.2 (1952), pp. 341–366. ISSN: 00029947.

References IV



T. Strohmer. "A note on equiangular tight frames". In: *Linear Algebra and its Applications* 429 (2008), pp. 326–330. DOI: 10.1016/j.laa.2008.02.030.



N. Weaver. "The Kadison-Singer problem in discrepancy theory". In: *Discrete Mathematics* 278 (Sept. 2004), pp. 227–239.